

# PRACTICAL

## WEEK-9

### 2520090046

## BOOKFLOW LIBRARY SYSTEM

Transactional Microservice using Spring Boot, Spring Data JPA and PostgreSQL

To develop a transactional BookFlow Library microservice using Spring Boot, Spring Data JPA and PostgreSQL for managing book inventory and borrowing orders, with FastAPI Gateway integration.

**Database – bookflow creation in Postgresql**

```
postgres=# CREATE DATABASE bookflow;
CREATE DATABASE
postgres=# \c bookflow
You are now connected to database "bookflow" as user "postgres".
bookflow=# \dt
Did not find any tables.
```

**Tables Inserted**

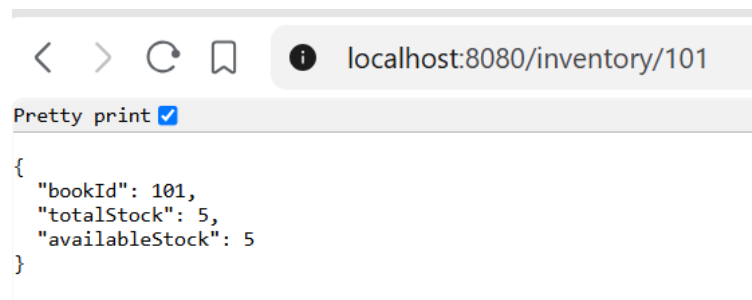
```
bookflow=# \dt
          List of tables
 Schema |   Name   | Type | Owner
-----+-----+-----+-----
 public | inventory | table | postgres
 public | orders    | table | postgres
(2 rows)
```

**Initial Inventory Data in PostgreSQL**

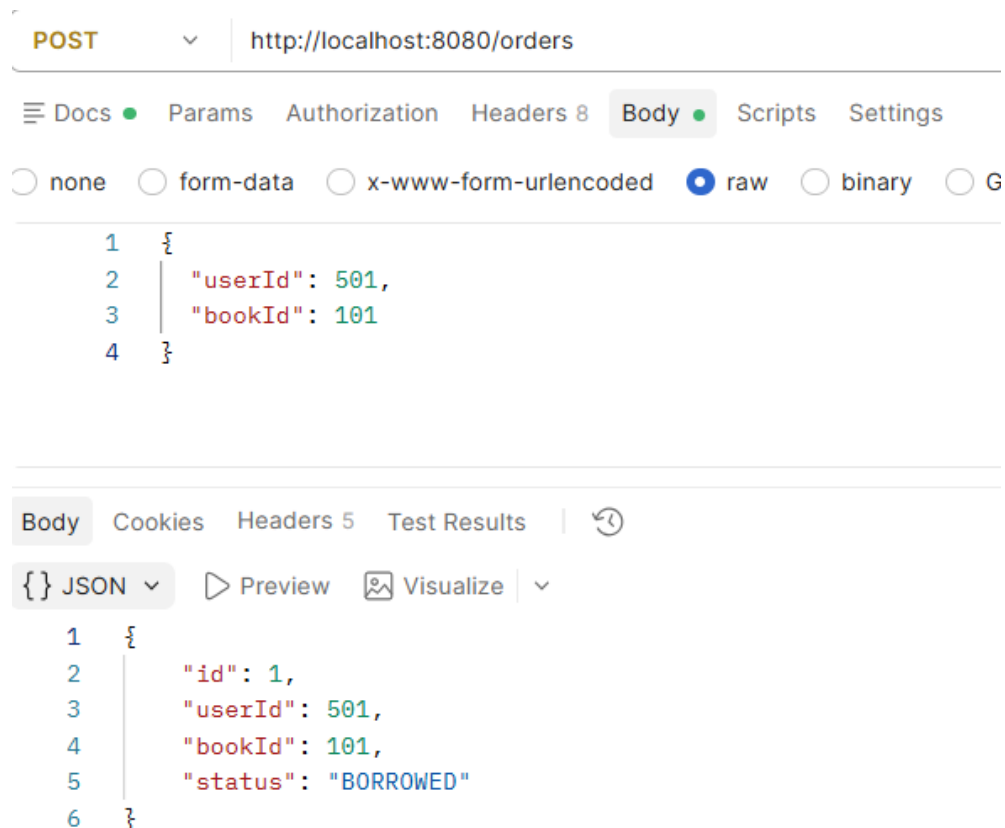
```
bookflow=# INSERT INTO inventory
bookflow-# (book_id, total_stock, available_stock)
bookflow-# VALUES
bookflow-# (101, 5, 5),
bookflow-# (102, 3, 3),
bookflow-# (103, 2, 0);
INSERT 0 3
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
    101 |          5     |          5
    102 |          3     |          3
    103 |          0     |          2
(3 rows)
```

## GET Inventory for Book 101

```
2026-10-01T12:02:30.250+05:30 INFO 23396 --- [demo] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
Hibernate:
    select
        i1_0.book_id,
        i1_0.available_stock,
        i1_0.total_stock
    from
        inventory i1_0
    where
        i1_0.book_id=?
```



## Successful Book Borrowing — POST /orders



## GET /inventory/101 — Inventory After Book Borrowing

```
Hibernate:
  select
    i1_0.book_id,
    i1_0.available_stock,
    i1_0.total_stock
  from
    inventory i1_0
  where
    i1_0.book_id=?
```

localhost:8080/inventory/101

Pretty print ☒

```
{
  "bookId": 101,
  "totalStock": 5,
  "availableStock": 4
}
```

## PostgreSQL Database — Inventory and Order Records After Borrowing

```
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock 
-----+-----+-----
    101  |             5  |          5
    102  |             3  |          3
    103  |             0  |          2
(3 rows)
```

```
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock 
-----+-----+-----
    102  |             3  |          3
    103  |             0  |          2
    101  |             4  |          5
(3 rows)
```

```
bookflow=# SELECT * FROM orders;
 id | book_id | status  | user_id 
----+-----+-----+-----
  1 |    101 | BORROWED |     501
(1 row)
```

## POST /orders — Out-of-Stock Book Returns 409 Conflict

The screenshot shows a Postman interface for a workspace named "M SAMPATH's Workspace". The active tab is "Event & Room Occupancy Management System > Health > Health Check". A POST request is configured to "http://localhost:8080/orders" with a JSON body: 

```
{  "userId": 502,  "bookId": 103}
```

. The "Body" tab is selected, showing a note: "Book out of stock". The "Test Results" tab is also visible, displaying the following JSON response: 

```
{  "message": "Book 103 is out of stock",  "status": 409,  "error": "OUT_OF_STOCK"}
```

## FastAPI Gateway — /gateway/orders Successfully Forwards Request to Spring Boot

**INFO:** 127.0.0.1:64794 - "POST /gateway/orders HTTP/1.1" 200 OK

The screenshot shows the same Postman interface, but the request is now to "http://localhost:8000/gateway/orders". The JSON body is: 

```
{  "userId": 501,  "bookId": 101}
```

. The "Test Results" tab displays the following JSON response: 

```
{  "id": 2,  "userId": 501,  "bookId": 101,  "status": "BORROWED"}
```

## PostgreSQL — Final Inventory and Order Records

```
bookflow=# SELECT * FROM inventory;
 book_id | available_stock | total_stock
-----+-----+-----
      102 |                3 |          3
      103 |                0 |          2
      101 |                3 |          5
(3 rows)
```

```
bookflow=# SELECT * FROM orders;
 id | book_id | status  | user_id
----+-----+-----+-----
   1 |     101 | BORROWED |     501
   2 |     101 | BORROWED |     501
(2 rows)
```

### Folder Structure:

bookflow-transaction/

```
|
|
| — pom.xml
|
|
└─ src/
    └─ main/
        └─ java/
            └─ com/
                └─ bookflow/
                    └─ transaction/
                        |
                        | — BookflowTransactionApplication.java
                        |
                        | — controller/
                        |   └─ OrderController.java
                        |   └─ InventoryController.java
                        |
                        |
```

```

|   |   └─ entity/
|   |   |   └─ Inventory.java
|   |   |   └─ Order.java
|   |   |
|   |   └─ repository/
|   |   |   └─ InventoryRepository.java
|   |   |   └─ OrderRepository.java
|   |   |
|   |   └─ service/
|   |   |   └─ InventoryService.java
|   |   |   └─ OrderService.java
|   |   |
|   |   └─ dto/
|   |   |   └─ OrderRequest.java
|   |   |
|   |   └─ exception/
|   |   |   └─ BookOutOfStockException.java
|   |   |   └─ GlobalExceptionHandler.java
|   |
└─ resources/
    └─ application.yml

```

## Codes:

- pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.5</version>

```

```
        <relativePath/>
    </parent>

    <groupId>com.bookflow</groupId>
    <artifactId>bookflow-transaction</artifactId>
    <version>0.0.1-SNAPSHOT</version>

    <name>bookflow-transaction</name>
    <description>BookFlow Transactional Microservice</description>

    <properties>
        <java.version>17</java.version>
    </properties>

    <dependencies>

        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>

        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-data-jpa</artifactId>
        </dependency>

        <dependency>
            <groupId>org.postgresql</groupId>
            <artifactId>postgresql</artifactId>
            <scope>runtime</scope>
        </dependency>

        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-validation</artifactId>
        </dependency>

        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-test</artifactId>
            <scope>test</scope>
        </dependency>

    </dependencies>

    <build>
        <plugins>
            <plugin>
```

```

        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
        <configuration>
            <mainClass>com.bookflow.transaction.BookflowTransactionApplication</mainClass>
        </configuration>
    </plugin>
</plugins>
</build>

</project>

```

- application.yml

```

spring:
  datasource:
    url: jdbc:postgresql://localhost:5432/bookflow
    username: postgres
    password: UR_PASSWORD

  jpa:
    hibernate:
      ddl-auto: update
      show-sql: true
    properties:
      hibernate:
        format_sql: true

server:
  port: 8080

```

- BookflowTransactionApplication.java

```

package com.bookflow.transaction;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class BookflowTransactionApplication {

    public static void main(String[] args) {
        SpringApplication.run(BookflowTransactionApplication.class, args);
    }
}

```

- Inventory.java

```
package com.bookflow.transaction.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import jakarta.validation.constraints.Min;

@Entity
@Table(name = "inventory")
public class Inventory {

    @Id
    private Long bookId;

    @Min(value = 0, message = "Total stock cannot be negative")
    private Integer totalStock;

    @Min(value = 0, message = "Available stock cannot be negative")
    private Integer availableStock;

    public Inventory() {
    }

    public Inventory(Long bookId, Integer totalStock, Integer availableStock)
    {
        this.bookId = bookId;
        this.totalStock = totalStock;
        this.availableStock = availableStock;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }

    public Integer getTotalStock() {
        return totalStock;
    }

    public void setTotalStock(Integer totalStock) {
        this.totalStock = totalStock;
    }
}
```

```

    public Integer getAvailableStock() {
        return availableStock;
    }

    public void setAvailableStock(Integer availableStock) {
        this.availableStock = availableStock;
    }
}

```

- Order.java

```

package com.bookflow.transaction.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "orders")
public class Order {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private Long userId;
    private Long bookId;
    private String status;

    public Order() {
    }

    public Order(Long userId, Long bookId, String status) {
        this.userId = userId;
        this.bookId = bookId;
        this.status = status;
    }

    public Long getId() {
        return id;
    }

    public void setId(Long id) {
        this.id = id;
    }
}

```

```

    public Long getUserId() {
        return userId;
    }

    public void setUserId(Long userId) {
        this.userId = userId;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}

```

- InventoryRepository.java

```

package com.bookflow.transaction.repository;

import com.bookflow.transaction.entity.Inventory;
import org.springframework.data.jpa.repository.JpaRepository;

public interface InventoryRepository extends JpaRepository<Inventory, Long> {
}

```

- OrderRepository.java

```

package com.bookflow.transaction.repository;

import com.bookflow.transaction.entity.Order;
import org.springframework.data.jpa.repository.JpaRepository;

public interface OrderRepository extends JpaRepository<Order, Long> {
}

```

- OrderRequest.java

```
package com.bookflow.transaction.dto;

import jakarta.validation.constraints.NotNull;

public class OrderRequest {

    @NotNull
    private Long userId;

    @NotNull
    private Long bookId;

    public Long getUserId() {
        return userId;
    }

    public void setUserId(Long userId) {
        this.userId = userId;
    }

    public Long getBookId() {
        return bookId;
    }

    public void setBookId(Long bookId) {
        this.bookId = bookId;
    }
}
```

- InventoryService.java

```
package com.bookflow.transaction.service;

import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.repository.InventoryRepository;
import org.springframework.stereotype.Service;

@Service
public class InventoryService {

    private final InventoryRepository inventoryRepository;

    public InventoryService(InventoryRepository inventoryRepository) {
        this.inventoryRepository = inventoryRepository;
    }
}
```

```

        public Inventory getInventory(Long bookId) {
            return inventoryRepository.findById(bookId)
                .orElseThrow(() -> new RuntimeException("Book not found"));
        }
    }
}

```

- OrderService.java

```

package com.bookflow.transaction.service;

import com.bookflow.transaction.dto.OrderRequest;
import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.entity.Order;
import com.bookflow.transaction.exception.BookOutOfStockException;
import com.bookflow.transaction.repository.InventoryRepository;
import com.bookflow.transaction.repository.OrderRepository;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
public class OrderService {

    private final InventoryRepository inventoryRepository;
    private final OrderRepository orderRepository;

    public OrderService(
        InventoryRepository inventoryRepository,
        OrderRepository orderRepository) {
        this.inventoryRepository = inventoryRepository;
        this.orderRepository = orderRepository;
    }

    @Transactional
    public Order borrowBook(OrderRequest request) {

        Inventory inventory = inventoryRepository.findById(request.getBookId())
            .orElseThrow(() -> new RuntimeException("Book not found"));

        if (inventory.getAvailableStock() <= 0) {
            throw new BookOutOfStockException(
                "Book " + request.getBookId() + " is out of stock");
        }

        inventory.setAvailableStock(
            inventory.getAvailableStock() - 1);

        inventoryRepository.save(inventory);
    }
}

```

```

        Order order = new Order(
            request.getUserId(),
            request.getBookId(),
            "BORROWED");

        return orderRepository.save(order);
    }
}

```

- InventoryController.java

```

package com.bookflow.transaction.controller;

import com.bookflow.transaction.entity.Inventory;
import com.bookflow.transaction.service.InventoryService;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/inventory")
public class InventoryController {

    private final InventoryService inventoryService;

    public InventoryController(InventoryService inventoryService) {
        this.inventoryService = inventoryService;
    }

    @GetMapping("/{bookId}")
    public Inventory getInventory(@PathVariable Long bookId) {
        return inventoryService.getInventory(bookId);
    }
}

```

- OrderController.java

```

package com.bookflow.transaction.controller;

import com.bookflow.transaction.dto.OrderRequest;
import com.bookflow.transaction.entity.Order;
import com.bookflow.transaction.service.OrderService;
import jakarta.validation.Valid;
import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/orders")
public class OrderController {

```

```

private final OrderService orderService;

public OrderController(OrderService orderService) {
    this.orderService = orderService;
}

@PostMapping
@ResponseStatus(HttpStatus.CREATED)
public Order borrowBook(@Valid @RequestBody OrderRequest request) {
    return orderService.borrowBook(request);
}
}

```

- BookOutOfStockException.java

```

package com.bookflow.transaction.exception;

public class BookOutOfStockException extends RuntimeException {

    public BookOutOfStockException(String message) {
        super(message);
    }
}

```

- GlobalExceptionHandler.java

```

package com.bookflow.transaction.exception;

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;

import java.util.Map;

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(BookOutOfStockException.class)
    @ResponseStatus(HttpStatus.CONFLICT)
    public Map<String, Object> handleOutOfStock(
        BookOutOfStockException ex) {

        return Map.of(
            "status", 409,
            "error", "OUT_OF_STOCK",
            "message", ex.getMessage()
        );
    }
}

```

```
}
```

- gateway.py — FastAPI gateway

```
from fastapi import FastAPI
import httpx

app = FastAPI()

SPRING_SERVICE = "http://localhost:8080"

@app.post("/gateway/orders")
async def create_order(order: dict):
    async with httpx.AsyncClient() as client:
        response = await client.post(
            f"{SPRING_SERVICE}/orders",
            json=order
        )
    return response.json()
```